

»» Chapter III ««

TOPOLOGICAL AND METRIC SPACES

PREFATORY REMARKS

When we introduce topological and metric spaces in a first book on complex function theory, we may appear to be insisting on an excessive amount of generality. Of the many justifications a principal one is that in complex analysis we encounter again and again certain general ideas that are particularly important in achieving global, as distinguished from local, results; central among these ideas are *compactness* and *connectedness*. They serve as binding agents. The reader will have many occasions to see these two ideas put to work. Since no extra effort is needed to introduce them in a general setting and since it is sound pedagogy to have general principles wanted for many parts of modern mathematics thoroughly assimilated early, no further defense is needed for the level of generality employed in this chapter. The object is not, of course, to teach general topology while “standing on one foot,” but rather to present a useful *vade mecum* for the remainder of this book. For a detailed authoritative modern account of general topology cf. [65].

There is another side to the topological requirements for the worker in certain areas of one variable complex function theory; it is the topology of the plane and, more generally, of two-dimensional manifolds. The demands for a specialized knowledge of such material do not appear in the early

chapters of our subject. We shall not have to concern ourselves anywhere in the present book with such topics as the Jordan curve theorem.†

A very readable treatment of the theory of metric spaces and the topology of the plane is that of Newman [92].

For the student who has taken a standard undergraduate course on general topology this chapter will be little more than a review. Experience of the present time indicates that only a fraction of a beginning class in complex function theory has had a prior course on general topology.‡ The remainder of such a class, having had a course on elementary real analysis, should find it a simple matter to become familiar with the rudiments of general topology and of the theory of metric spaces, as presented here, save possibly for the notion of *connectedness* which the newcomer may find is not directly referable to his earlier experience. However, it is not more “difficult” than many concepts already encountered. By the study of examples in $\mathbf{R}$ and $\mathbf{C}$ the beginner should convince himself that the concept is “reasonable” and by the study of the role it plays that it is important.

A model for the modern exposition of general topology is the treatise of Bourbaki (cf., in particular, [19]). Its influence will be strongly felt in this chapter.

1. THE NOTION OF A TOPOLOGICAL SPACE

The student is familiar with the following notion of an *open* subset of $\mathbf{R}$: $O(\subset \mathbf{R})$ is said to be *open* provided that for each $a \in O$ there exists r positive such that $]a - r, a + r[\subset O$. It is to be observed that the definition distinguishes a family of subsets of $\mathbf{R}$, those that are open, and that the following useful properties hold: (a) given an arbitrary family Φ of open parts of $\mathbf{R}$, $\bigcup_{O \in \Phi} O$ is itself open; (b) if O_1 and O_2 are open parts of $\mathbf{R}$, so is $O_1 \cap O_2$; (c) $\emptyset$ and $\mathbf{R}$ are open; (d) given $a, b \in \mathbf{R}$, $a \neq b$, there exist open O_1 and O_2 such that $a \in O_1$, $b \in O_2$, $O_1 \cap O_2 = \emptyset$.

The verifications are routine. It is not hard to sense that the stated properties may be thought of as applying to the members of a distinguished family of subsets of a given set, reference to $\mathbf{R}$ being eliminated. Granting on the basis of experience that the properties (a)–(d) inclusive are primordial in the study of questions involving open subsets of $\mathbf{R}$, we have reasonable grounds to suppose that one can isolate useful general working principles by dealing with the question in an abstract setting.

† To the interested reader we recommend F. Riesz’ beautiful proof of the Jordan curve theorem which is fully accessible by the end of §5, Ch. V of this book. For an account in English cf. [53].

‡ Perhaps the preceding sentence will soon be obsolete!

We start with a set X . Let $T \subset \mathfrak{P}(X)$. We say that T is a *topology* on X provided that the counterparts of (a), (b), and (c) hold for the subsets of X which are members of T , that is, provided that T satisfies: (a) if $\Phi \subset T$, then $\bigcup_{O \in \Phi} O \in T$; (b) given $O_1, O_2 \in T$, then $O_1 \cap O_2 \in T$; (c) $\emptyset, X \in T$. T is said to be a *Hausdorff topology* provided that in addition T satisfies: (d) given $a, b \in X, a \neq b$, there exist $O_1, O_2 \in T$ such that $a \in O_1, b \in O_2, O_1 \cap O_2 = \emptyset$. The composite (X, T) of a set X together with a topology T on X is termed a *topological space*; it is termed a *Hausdorff (topological) space* provided T is a Hausdorff topology. The members of T are termed *open sets*. We speak loosely of X as a topological space. The elements of X are termed the *points* of X .

Given $S \subset \mathfrak{P}(X)$, the intersection of the topologies containing S is itself a topology, $T(S)$, and it contains S . It is the smallest topology containing S (that is, it is contained in every topology containing S). $T(S)$ is termed the *topology on X generated by S* .

EXERCISES

1.1. The topology on $\mathbf{R}$ given in the first paragraph of this section (T = family of open sets with "open" as defined there) is generated by the family of intervals $]a, b[, a, b \in \mathbf{R}, a < b$. Every open set is the union of a family of intervals.

We term this topology the *standard topology* or more simply the *topology* on $\mathbf{R}$. Of course there are other topologies on $\mathbf{R}$ [for example, $\mathfrak{P}(\mathbf{R})$], however we shall use only the standard topology.

1.2. By the *standard topology* (or more simply the *topology*) on $\mathbf{C}$ is meant the topology generated by the family of circular disks $\Delta(a; r), a \in \mathbf{C}, r$ positive. Show that it is a Hausdorff topology and that $A(\subset \mathbf{C})$ is open in the sense of this topology if and only if for each $a \in A$ there exists r positive such that $\Delta(a; r) \subset A$ (equivalently, if and only if A is the union of a family of disks).

We shall use only the standard topology on $\mathbf{C}$.

1.3. Given $a \in \mathbf{R}$, the following subsets of $\mathbf{C}$ are open: $\{\operatorname{Re} z > a\}, \{\operatorname{Re} z < a\}, \{\operatorname{Im} z > a\}, \{\operatorname{Im} z < a\}, \{|z| > a\}$.

Continuing with a topological space (X, T) we introduce several notions that occur commonly. A set $F \subset X$ is termed *closed* provided that $X - F$ is open. Given $x \in X$, a set $V \subset X$ is said to be a *neighborhood* of x provided that there exists $O \in T$ such that $x \in O \subset V$. We note that $\Delta(a; r)$ and $\{|z - a| \leq r\}, r$ positive, are neighborhoods of a . We note also that old usage referred only to special kinds of sets as neighborhoods (intervals, disks, ...). The modern usage is that of the definition. Given a set $A \subset X$ and $a \in X$, we say that a *adheres* to A (or a is a *contact point* of A) provided that $V \cap A \neq \emptyset$ for each neighborhood V of a . By the *adherence* or *closure* of A , denoted $\bar{A}$, is meant the set of points that adhere to A . By the *frontier* of A , denoted $\operatorname{fr} A$, is meant $\bar{A} \cap \overline{X - A}$. We say that A *clusters at a* provided that $a \in \bar{A} - \{a\}$.

EXERCISES

- 1.4. Given $A \subset X$, $\bar{A}$ is closed. A is closed if and only if $A = \bar{A}$.
 1.5. If A is open, $\text{fr } A = \bar{A} - A$.
 1.6. In $\mathbb{C}$ we have: $\overline{\Delta(a; r)} = \{|z - a| \leq r\}$, $\text{fr } \Delta(a; r) = C(a; r)$.

Given topological spaces X and Y , each with its own topology. Let $f: X \rightarrow Y$ and $x \in X$. We say that f is *continuous at x* provided that for each neighborhood V of $f(x)$ there exists a neighborhood U of x such that $f(U) \subset V$, or, equivalently, $f(y) \in V$, $y \in U$.

EXERCISE

- 1.7. Show that f is continuous at x if and only if $f^{-1}(V)$ is a neighborhood of x whenever V is a neighborhood of $f(x)$.

We say that f is *continuous on X* (or that f is a *continuous map of X into Y*) provided that f is continuous at each point of X .†

EXERCISES

- 1.8. f is continuous on X if and only if for each O open in Y (that is, a member of the topology on Y) $f^{-1}(O)$ is open in X .
 1.9. Composition. Suppose that X, Y, Z are topological spaces, $f: X \rightarrow Y, g: Y \rightarrow Z$, f is continuous at $x \in X$, g is continuous at $f(x)$. Then $g \circ f$ is continuous at x .

Relative topology. Given a topological space (X, T) and $A \subset X$. The family of sets $O \cap A$, $O \in T$, constitutes a topology T_A on A , which is a Hausdorff topology when T is. It is termed the *relative topology on A* . Note that the standard topology on $\mathbb{R}$ is the relative topology on $\mathbb{R}$ in the sense of $\mathbb{C}$. If now $f: A \rightarrow Y$ where Y is a topological space, we say that f is *continuous at $a \in A$* (resp., *on A*) provided that f is continuous at a (on A) in the sense of T_A . The consistency is obvious when $A = X$.

EXERCISES

- 1.10. Check that for the case $A \subset X = \mathbb{R}$, $Y = \mathbb{R}$, the definition of continuity just given is equivalent to the usual “ ε, δ ” definition. The same holds if $\mathbb{R}$ is replaced by $\mathbb{C}$ for either X or Y .

† “Onto” replaces “into” when it is known that $Y = f(X)$.

1.11. Given X, Y are topological spaces, $A \subset X, f: A \rightarrow Y$. Then f is continuous on A if and only if $f: A \rightarrow f(A)$ is continuous on A in the sense of the relative topologies of A and $f(A)$.

Homeomorphism. Given topological spaces (X_1, T_1) and (X_2, T_2) we say that f is a *homeomorphism* of X_1 onto X_2 provided that f is a univalent map of X_1 onto X_2 such that the map with domain T_1 taking a set O into $f(O)$ has image T_2 . It is to be noticed that if f is a homeomorphism of X_1 onto X_2 , then both f and f^{-1} are continuous on their domains by virtue of Ex. 1.8 of this section. The converse is also true: if f is a univalent map of X_1 onto X_2 such that f and f^{-1} are continuous on their respective domains, then f is a homeomorphism of X_1 onto X_2 . Indeed, $f(O) \in T_2$ for $O \in T_1$ and $f^{-1}(\Omega) \in T_1$ for $\Omega \in T_2$, so that the map taking O into $f(O)$ maps T_1 onto T_2 . Topology is concerned with properties preserved under homeomorphisms.

EXERCISE

1.12. Show that adherence and frontier are preserved under homeomorphism, that is, if f is a homeomorphism, then $f(\bar{A}) = \overline{f(A)}$, $f(\text{fr } A) = \text{fr}[f(A)]$.

Limit. Given topological spaces X and Y , the latter being Hausdorff. Let $a \in \overline{X - \{a\}}$ and let $f: A \rightarrow Y$ where A is $X - \{a\}$ or X . By the Hausdorff property there is at most one element $b \in Y$ such that $f|(X - \{a\}) \cup (a, b)$ is continuous at a . When such a b exists, we say that f has *limit* b at a and denote b by $\lim_a f$. Once more we note that our conventions on the employment of the limit notation are to prevail; implicit in its use without qualification is the existence of a limit.

Limit of a sequence. Given $(b_n): \mathbb{N} \rightarrow Y$, there is at most one element $b \in Y$ such that for each neighborhood V of b there exists $m \in \mathbb{N}$ such that $b_n \in V$, $n \geq m$. When such b exists, we say that (b_n) has *limit* b and denote b as in the earlier situations, by $\lim b_n$.†

Product topology. Given topological spaces (X_1, T_1) and (X_2, T_2) we may endow $X_1 \times X_2$ with a useful topology as follows. Let T denote the family of unions of sets of the form $O_1 \times O_2$, $(O_1, O_2) \in T_1 \times T_2$. It is readily verified that T is a topology on $X_1 \times X_2$. T is called the *product topology* of $X_1 \times X_2$. When X_1 and X_2 are Hausdorff, so is $X_1 \times X_2$. Further the projection map carrying (x_1, x_2) into x_k is a continuous map of $X_1 \times X_2$ onto

† We shall see in §6, this chapter, that the notion of the limit of a sequence may be subsumed under that of a map once we introduce the extended real line.

$X_k, k = 1, 2$, when X_1 and X_2 are not empty (and vacuously when X_1 and X_2 are empty).

EXERCISE

1.13. Verify the assertions of the preceding paragraph.

2. COMPACTNESS

Given a set $X, \Phi \subset \mathfrak{P}(X)$ is said to be a *covering* of X provided that $X = \bigcup_{E \in \Phi} E$. Ψ is said to be a Φ *subcovering* provided that $\Psi \subset \Phi$ and Ψ is a covering of X . When (X, T) is a topological space, Φ is said to be an *open covering* of X provided that Φ is a T subcovering.

A topological space X is said to be *compact* provided that X is a Hausdorff space and that the Heine–Borel–Lebesgue property holds: *for each open covering Ω of X , there exists a finite Ω subcovering*. The value of the notion of compactness is to be found in its fundamental properties and in the fact that a number of important spaces are compact.

A set A contained in a Hausdorff space X is said to be *compact* provided that A is a compact space in the sense of its relative topology. This definition is consistent when $A = X$. It is easy to see that $A \subset X$ is compact if and only if whenever $\Omega \subset T$ has the property $A \subset \bigcup_{O \in \Omega} O$, there exists Φ finite $\subset \Omega$ such that $A \subset \bigcup_{O \in \Phi} O$. A second simple fact concerning compact sets is: *a compact set is closed*. Indeed, suppose that A is a compact set in X . We show that $X - A$ is open. We suppose that $A \neq \emptyset, X$, for otherwise the assertion would be obvious. Let $b \in X - A$ and note that by the Hausdorff property for each $a \in A$ there exist open sets $\omega(a), O(a)$ of X satisfying $a \in \omega(a), b \in O(a), \omega(a) \cap O(a) = \emptyset$. By the compactness of A there exists a finite set $\{\omega(a_1), \dots, \omega(a_n)\}$ such that $A \subset \bigcup_{1 \leq k \leq n} \omega(a_k)$. Now

$$b \in \bigcap_{1 \leq k \leq n} O(a_k) \subset X - A$$

and the intersection of the $O(a_k)$ is open. Consequently, $X - A$ is representable as the union of a family of open sets and hence is open. The assertion follows.

A useful fact concerning compactness is: *a closed subset A of a compact space X is compact*. Let $\Omega \subset T$ and $A \subset \bigcup_{O \in \Omega} O$. Then $\Omega \cup \{X - A\}$ is an open covering of X . Consequently, some finite subset Φ of Ω has the property that $X = (X - A) \cup \bigcup_{O \in \Phi} O$ and hence $A \subset \bigcup_{O \in \Phi} O$.† An immediate

† We are using the fact that a subset of a finite set is finite.

corollary is: If $A \subset B \subset X$, where X is a Hausdorff space, B is compact, and A is a closed subset of X , then A is compact.

The action of a continuous map on a compact set is of special interest. Here one must sharply distinguish "closed" and "compact." A topological space is always closed in the sense of its topology, its image with respect to a continuous map need not be. For example, let $\iota: [0, 1[\rightarrow \mathbf{R}$, $\iota(x) = x$, the topology on $[0, 1[$ being the relative topology. The image is not closed in $\mathbf{R}$. Even for such simple spaces as $\mathbf{R}$ the image need not be closed. Consider $f(x) = x(1 + |x|)^{-1}$, $x \in \mathbf{R}$. For compact spaces the situation is more favorable. We have

Theorem 2.1: *Let $f: X \rightarrow Y$ where X is a compact space, Y is a Hausdorff space, and f is continuous on X . Then $f(X)$ is compact.*

PROOF: We put aside the trivial case where $X = \emptyset$. Let Ω denote a set of open sets in Y with union containing $f(X)$. Then $\{f^{-1}(O): O \in \Omega\}$ is an open covering of X . By the compactness of X , there is a finite subcovering $\{f^{-1}(O_1), \dots, f^{-1}(O_n)\}$ and we have $f(X) \subset \bigcup_{1 \leq k \leq n} O_k$. Hence $f(X)$ is compact.

EXERCISE

2.1. Given $f: X \rightarrow Y$ where X is compact, Y is Hausdorff, and f is univalent and continuous on X . Show that f is a homeomorphism of X onto $f(X)$. Hint: Show that if $O \subset X$ is open, then $f(O)$ is open in the sense of the relative topology of $f(X)$.

From the point of view of complex function theory interest attaches to the compact subsets of $\mathbf{R}$ and $\mathbf{C}$. Suppose that A is a compact subset of $\mathbf{R}$. Since the set of intervals $] -x, x[$, x positive, is an open covering of A , $A \subset] -x, x[$ for some positive x . We conclude that A is bounded and we have already seen that A is closed. The same statements hold for a compact subset of $\mathbf{C}$.

The converse statements hold. We shall now prove: *If A is a bounded closed subset of $\mathbf{R}$, then A is compact* (Theorem of Heine–Borel–Lebesgue). It suffices to prove the assertion when A is a segment, a bounded closed set of $\mathbf{R}$ being a closed set contained in a segment and hence compact, granted that a segment is. The compactness of a segment may be established in many different ways. On purely practical grounds we choose the *method of successive bisections* because it is the prototype of the method we shall want for the Cauchy–Goursat theorem, §2, Ch. V.

PROOF: Let $s = [a, b]$, $a \leq b$, be the given segment. Let Ω denote a set of open sets in $\mathbf{R}$ whose union contains $[a, b]$. Suppose that there does not exist Φ finite $\subset \Omega$ such that $s \subset \bigcup_{O \in \Phi} O$. Now $s = [a, (a+b)/2] \cup [(a+b)/2, b]$ and at least one of the segments on the right-hand side, which are termed the *bisegments* of s , has the property (f): there does not exist Φ finite $\subset \Omega$ such that it is contained in $\bigcup_{O \in \Phi} O$.† If $[a, (a+b)/2]$ has the property (f), we denote it by s^1 , and otherwise we denote $[(a+b)/2, b]$ by s^1 . We conclude inductively that there exists a sequence $(s_n)_{n \in \mathbf{N}}$: $s_0 = s$, s_{n+1} is a bisegment of s_n , and s_n has the property (f), $n \in \mathbf{N}$. In fact, we may construct $(s_n)_{n \in \mathbf{N}}$ by taking the union of $(s_n)_0^m$, $m \in \mathbf{N}$, having the property that $s_0 = s$; each s_n has the property (f), $n = 0, \dots, m$; and s_{n+1} is obtained from s_n as s^1 is from s . It is to be noted that the $(s_n)_0^m$ are unique and exist for all $m \in \mathbf{N}$, $(s_n)_0^{m+1}$ being $(s_n)_0^m \cup \{(m+1, s_{m+1})\}$ where s_{m+1} is obtained from s_m as s^1 is from s . Since $s_{n+1} \subset s_n$, $n \in \mathbf{N}$, the left (right) endpoint of s_n is the n th term of a monotone never-decreasing (never-increasing) sequence which is bounded above (below). Let c denote the limit of the sequence associated with the left endpoints and d the limit of the sequence associated with the right endpoints. Since $[c, d] \subset s_n$, $n \in \mathbf{N}$, we have $0 \leq d - c \leq 2^{-n}(b - a)$, $n \in \mathbf{N}$. Hence $d = c$. Now $c \in O \in \Omega$ and $s_n \subset O$ for n large, so that for such n , s_n does not have the property (f). Contradiction. Consequently, there exists Φ finite $\subset \Omega$ such that $s \subset \bigcup_{O \in \Phi} O$.

EXERCISE

2.2. Let $a, b, c, d \in \mathbf{R}$, $a \leq b$ and $c \leq d$. Show that $\sigma([a, b] \times [c, d])$ is compact.

When $A \subset \mathbf{R}$ is compact and not empty, then A possesses both a minimum and a maximum. In fact, $\inf A$, $\sup A \in \bar{A}$, and $\bar{A} = A$ since A is closed. We are led to a fundamental conclusion (Weierstrass maximum and minimum theorem for continuous functions).

Theorem 2.2: Given $f: X \rightarrow \mathbf{R}$ where X is a nonempty compact space and f is continuous on X . Then $f(X)$ possesses a maximum and a minimum.

Although we shall not consider semicontinuity further in this book, it is appropriate to point out, because of the importance of the notion, that Theorem 2.2 is capable of extension to the case of semicontinuous functions. A function $f: X \rightarrow \mathbf{R}$ where X is a topological space is said to be *lower semicontinuous* at $a \in X$ provided that for each real $b < f(a)$, $f^{-1}(\{x > b\})$ is a

† “f” = “failure.” We are using the fact that $A \cup B$ is finite when A and B are.

neighborhood of a ; f is said to be *lower semicontinuous on X* provided that f is lower semicontinuous at each point of X . If f is lower semicontinuous on X , the sets $f^{-1}(\{x > b\})$, $b \in \mathbb{R}$, constitute an open covering of X ; whence if X is compact and not empty, $f(X)$ has an infimum β . Further β must be the *minimum* of $f(X)$. For otherwise the sets $f^{-1}(\{x > b\})$, $b > \beta$, would constitute an open covering of X and a second application of compactness yields that $X = f^{-1}(\{x > b\})$ for some $b > \beta$, so that $\inf f(X) \geq b$. Contradiction. Corresponding statements hold for upper semicontinuity for which the definitions are given by reversing the inequalities in the above definitions.

EXERCISES

2.3. Let $n \in \mathbb{N}$, $a_k \in \mathbb{C}$, $k = 0, \dots, n$. Let $p(z) = z^{n+1} + \sum_0^n a_k z^k$, $z \in \mathbb{C}$. Show that $|p|$ has a minimum. Hint: Using $p(z) = z^{n+1}[1 + \sum_0^n a_k z^{k-(n+1)}]$ show that for some positive r we have $|p(z)| \geq |z|^{n+1}/2 \geq |p(0)|$, $|z| \geq r$.

2.4. Apply the method of Hurwitz (Ex. 5.9, Ch. II) to $f(z) = |\sum_0^n b_k z^k|^2$, considering $\lim(n+1)^{-1} \sum_0^n f(u^k z)$, to show that if $|\sum_0^n b_k z^k| \leq M$, $|z| = r$, then $\sum_0^n |b_k|^2 r^{2k} \leq M^2$ and thereupon conclude the corresponding inequality for a power series $\sum_0^\infty b_k z^k$, namely if $|\sum_0^\infty b_k z^k| \leq M$, $|z| = r$, then $\sum_0^\infty |b_k|^2 r^{2k} \leq M^2$, it being assumed that the radius of convergence is $+\infty$ or is positive but greater than r . Conclude that if $M = |b_0|$ and r is positive, then $b_k = 0$, $k \geq 1$.

2.5. Returning to the situation of Ex. 2.3 show that if $|p(c)| = \min\{|p(z)| : z \in \mathbb{C}\}$, then $p(c) = 0$, thereby establishing the "Fundamental" Theorem of Algebra. Hint: Suppose $p(c) \neq 0$ and apply Ex. 2.4 to $1/p(z+c)$, z small.

Prove as a corollary: If $\sum_0^\infty a_k z^k$ has radius of convergence $+\infty$ and $\{k : a_k \neq 0\}$ is unbounded, then the sum of $\sum_0^\infty a_k z^k$ on $\mathbb{C}$ is not representable as the quotient of polynomials.

3. CONNECTEDNESS

We continue with a topological space X not necessarily Hausdorff. By a *splitting* of X is meant a pair $\{O_1, O_2\}$ where (1) O_1, O_2 are open and non-empty, (2) $O_1 \cap O_2 = \emptyset$, and (3) $O_1 \cup O_2 = X$. We say that X is *connected* provided that there does not exist a splitting of X . Given $A \subset X$, we say that A is *connected* provided that A is a connected topological space in the sense of its relative topology. The importance of the notion will unfold. A premonition may be had by observing that if we are concerned with a sentential function $P(x)$, $x \in X$, where X is a connected topological space and we know that the sets $\{P(x)\}$ and $\{\text{not } P(x)\}$ are *both open*, then one is empty, so that either $P(x)$ holds for all $x \in X$ or else $P(x)$ holds for no $x \in X$. This observation yields a method of proof, termed the *standard connectedness argument* (cf. Theorem 4.2, this chapter). By a *region* we understand a nonempty open connected subset of a topological space. By a *continuum* we understand a

closed connected subset of a topological space which contains more than one point.

We show

Theorem 3.1: *Given $f: X \rightarrow Y$ where X and Y are topological spaces, X is connected, f is continuous on X . Then $f(X)$ is connected.*

PROOF: We assume, as we may, that $f(X) = Y$. Suppose that Y is not connected and that $\{O_1, O_2\}$ is a splitting of Y . Then $\{f^{-1}(O_1), f^{-1}(O_2)\}$ is a splitting of X . Contradiction.

Connected subsets of $\mathbf{R}$. From the point of view of classical analysis it is essential to have precise information on the connected subsets of $\mathbf{R}$. As we shall see, it is not hard to give a complete answer. The situation in $\mathbf{C}$ is far more complicated. Suppose then that $A \subset \mathbf{R}$ is connected and that $a, b \in A$, $a < b$. Suppose that $c \in]a, b[\cap (\mathbf{R} - A)$. Then $\{A \cap \{x < c\}, A \cap \{x > c\}\}$ is a splitting of A . This is impossible. Hence $[a, b] \subset A$. If $\sup A = +\infty$, then $\{x \geq c\} \subset A$, $c \in A$. Similarly, if $\inf A = -\infty$, then $\{x \leq c\} \subset A$, $c \in A$. If A is bounded above and is not empty, then $\{c \leq x < \sup A\} \subset A$, $c \in A$, a corresponding statement holding for A bounded below. On considering the various possibilities we see that a nonempty connected subset of $\mathbf{R}$ must be a set of one of the following kinds: $\mathbf{R}$, $\{a < x\}$, $\{a \leq x\}$, $\{a > x\}$, $\{a \geq x\}$, $\{a\}$, $]a, b[$, $[a, b[$, $]a, b]$, $[a, b]$, where $a, b \in \mathbf{R}$, $a < b$. However, as yet we do not know whether, apart from $\{a\}$, any of the listed sets is connected, the connectedness of the singleton being trivial. The following general theorem will permit us to reduce the question to the case of a segment and we shall be able to conclude that *all the listed sets are connected*.

Theorem 3.2: *Let X be a topological space. Let $A_\lambda \subset X$, $\lambda \in \Lambda$, where each A_λ is connected and $\bigcap_{\lambda \in \Lambda} A_\lambda \neq \emptyset$. Then $\bigcup_{\lambda \in \Lambda} A_\lambda$ is connected.*

PROOF: We suppose, as we may, that $\Lambda \neq \emptyset$, the case where $\Lambda = \emptyset$ being trivial, and that $\bigcup_{\lambda \in \Lambda} A_\lambda = X$. Suppose that X is not connected and let $\{O_1, O_2\}$ be a splitting, the indexing being such that O_1 contains some point of $\bigcap_{\lambda \in \Lambda} A_\lambda$. Since A_λ is connected, $A_\lambda \cap O_2 = \emptyset$, $\lambda \in \Lambda$, and hence $O_2 = \emptyset$. Contradiction. The theorem follows.

We return to the listed subsets of $\mathbf{R}$. We note that

$$\mathbf{R} = \bigcup_{n \in \mathbf{N}} [-(n+1), n+1],$$

and

$$]a, b[= \bigcup_{n \in \mathbf{N}} [a + (n+m)^{-1}, b - (n+m)^{-1}]$$

where m is a sufficiently large positive integer. The representation of the remaining sets as unions of segments having a nonempty intersection is readily supplied and is left to the reader.

There remains to show that a *segment is connected*. Suppose that $[a, b]$ is not connected and that $\{O_1, O_2\}$ is a splitting such that $a \in O_1$. Let

$$c = \sup\{x: [a, x] \subset O_1\}.$$

Clearly, $a < c \leq b$. Suppose that $c < b$. If $c \in O_1$, then $[c - h, c + h] \subset O_1$ for sufficiently small positive h , and $[a, c + h] \subset O_1$ for such h . This is not possible. Hence $c \in O_2$ and consequently $[c - h, c + h] \subset O_2$ for some small positive h . For such h we have $[a, x] \not\subset O_1$ for $c - h \leq x$ and hence the supremum property of c is contradicted. Hence $c = b$, $[a, b[\subset O_1$, $\{b\} = O_2$. This is not possible, since O_2 contains more than one point. We conclude that a segment is connected.

The connected subsets of $\mathbf{R}$ are now under control.

EXERCISES

3.1. If A is a connected subset of a topological space X and $A \subset B \subset \bar{A}$, then B is connected.

3.2. Given $a, b \in \mathbf{R}$, $a < b$. Suppose that f is a continuous map of $[a, b]$ into $\mathbf{R}$. Show that $f([a, b])$ is a segment. Generalize.

3.3. Show that f of Ex. 3.2 is univalent if and only if it is increasing ($f(x) < f(y)$, $a \leq x < y \leq b$) or it is decreasing ($f(x) > f(y)$, $a \leq x < y \leq b$).

3.4. Suppose that $f: E \rightarrow \mathbf{R}$ where E is a connected subset of $\mathbf{R}$, f is univalent and is continuous on E . Show that f is a homeomorphism of E onto $f(E)$ and that, if O is an open subset of $\mathbf{R}$ which is contained in E , $f(O)$ is an open subset of $\mathbf{R}$. [We shall see that the maps one encounters in complex function theory, subject to obvious restrictions, have the property that they carry open subsets of $\mathbf{C}$ in their domain into open subsets of $\mathbf{C}$. The present exercise is intended to depict a prototype in the one real variable theory.]

Components. Given a topological space X . If $x \in X$, there is a unique maximal connected set, $C(x)$, containing x (maximal in the sense that it contains all connected sets in X of which x is an element). It suffices by virtue of Theorem 3.2 to take $C(x)$ as the union of all connected sets of X of which x is an element. By Ex. 3.1, $C(x)$ is a closed subset of X . We term $C(x)$ the *component* of X containing x . Given $E \subset X$, we say that E is a *component* of X provided that there exists $x \in X$ such that $E = C(x)$. Given $x, y \in X$, then either $C(x) = C(y)$ or $C(x) \cap C(y) = \emptyset$. For if $C(x) \cap C(y) \neq \emptyset$, then $C(x) \cup C(y)$ is connected and $C(x)$ and $C(y)$ must both be equal to $C(x) \cup C(y)$ and hence to each other. The notion of component is applied to $A \subset X$, A being taken with its relative topology.

In the special case of a topological space X , each point of which is contained in a *connected* neighborhood, each component of X is open. The components are then regions. This is the situation for an open subset of $\mathbf{C}$ as we shall learn in the next section.

4. ARC

We define an *arc* of a topological space X as a continuous map γ of a segment $[a, b]$, $a < b$, of $\mathbf{R}$ into X ; $\gamma(a)$ is termed the *initial point* of γ and $\gamma(b)$ is termed the *terminal point* of γ . The term “*path*” is also used for “*arc*.” The map γ need not be univalent. Much license is customary in connection with the terminology pertaining to arcs and the term “*arc*” is applied to what should be called “*image of arc*.” We shall use γ to denote also the image of the map γ . The meaning will be clear from context, but the distinction is an essential one. Given arcs γ_1 and γ_2 , each having domain $[0, 1]$, the terminal point of γ_1 being the initial point of γ_2 , we define $\gamma_1 \cdot \gamma_2$ as the arc given by $\gamma(t) = \gamma_1(2t)$, $0 \leq t \leq \frac{1}{2}$, $= \gamma_2(2t - 1)$, $\frac{1}{2} \leq t \leq 1$ (*product arc*). The space X is termed *arcwise connected* provided that given $p, q \in X$ there exists an arc with initial point p and terminal point q .

The following useful theorem is very easy to show.

Theorem 4.1: *An arcwise connected topological space is connected.*

PROOF: Suppose that $\{O_1, O_2\}$ is a splitting of the space. Let $p \in O_1$, $q \in O_2$, and let γ denote an arc with initial point p and terminal point q . Then $\{\gamma^{-1}(O_1), \gamma^{-1}(O_2)\}$ is a splitting of the domain of γ . However a segment is connected. Contradiction.

For regions in $\mathbf{C}$ we have a converse.

Theorem 4.2: *A region in $\mathbf{C}$ is arcwise connected.*

PROOF: The argument that follows will be our first encounter with the standard connectedness argument. Let A be a region in $\mathbf{C}$ and let $p \in A$. Let B denote the set of q in A for which there exists an arc γ lying in A of which p is the initial point and q the terminal point. We show that B and $A - B$ are both open. Since $B \neq \emptyset$, it will follow that $B = A$. We note that if $b \in \Delta(a; r)$, then $[a, b] \subset \Delta(a; r)$. Let $\vec{ab}$ denote the map with domain $[0, 1]$ taking t into $a(1 - t) + bt$. Suppose now that $q \in B$. Let $\Delta(q; r) \subset A$. Let γ be an arc in A with initial point p , terminal point q , and domain $[0, 1]$. Then $\gamma \cdot \vec{qs}$ is an arc in A , $s \in \Delta(q; r)$. Hence $\Delta(q; r) \subset B$. Consequently, B is open.

Now if $q \in A - B$ and $s \in \Delta(q; r) \subset A$, then $s \in A - B$. For if $s \in B$, there exists an arc γ in A with initial point p , terminal point s , domain $[0, 1]$. Then $\gamma \cdot \overrightarrow{sq}$ is an arc in A , so that $q \in B$. Contradiction. We conclude that $A - B$ is open. Hence $B = A$. Since p is an arbitrary point of A , the theorem follows.

EXERCISE

4.1. Given $A \subset \mathbb{C}$, A convex. Show that A is arcwise connected. If

$$\Delta(a; r) \subset B \subset \overline{\Delta(a; r)},$$

then B is convex. Establish the final remark of §3, this chapter.

5. METRIC SPACE

Here we are concerned with a set on which a "reasonable" notion of distance is defined. Specifically, by a *metric space* is meant a set X together with a map $\rho: X \times X \rightarrow \mathbb{R}^+$ satisfying the following conditions: (1) $\rho(x, y) = 0$ if and only if $x = y$; (2) given $x, y \in X$, we have $\rho(x, y) = \rho(y, x)$ (symmetry); (3) given $x, y, z \in X$ we have $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$ (triangle inequality). We term $\rho(x, y)$ the (ρ -) *distance between x and y* and ρ a *metric on X* . Important cases for us are $\mathbb{R}$ and $\mathbb{C}$ with $\rho(x, y) = |x - y|$. The ρ so specified is termed "standard." By the *ball* with center $a \in X$, radius $r \in \mathbb{P}$, is meant $\{\rho(a, x) < r\}$. It is denoted $B(a; r)$.

Metric spaces serve as a convenient medium for treating uniform limits and uniform continuity. The degree of abstraction is not anything like the ultimate but is very adequate for our purposes.

However, before turning to these questions, we first consider a topology on X associated with ρ , and (quite independently) the notion of a *complete* metric space.

We ask for the topology T on a metric space X generated by the family of balls. It is termed the *topology on X induced by ρ* . We show that T consists of the unions of families of balls. To establish this assertion we note that if $c \in B(a; r) \cap B(b; s)$, then $B(c; t) \subset B(a; r) \cap B(b; s)$ when

$$t = \min\{r - \rho(a, c), s - \rho(b, c)\}.$$

Consequently, a point belonging to $A_1 \cap A_2$, where A_1 and A_2 are unions of balls, is the center of a ball contained in $A_1 \cap A_2$. Hence $A_1 \cap A_2$ is a union of balls. It is now immediate that the family of unions of balls is a topology on X and hence being contained in T is T . It now follows that a subset A of X is open if and only if for each $a \in A$ there exists r positive such that $B(a; r) \subset A$.

The topology T is a *Hausdorff* topology. Given $a, b \in X$, $a \neq b$, let $r = \rho(a, b)/2$. By the triangle inequality $B(a; r) \cap B(b; r) = \emptyset$.

EXERCISE

5.1. Consider the maps of $\mathbb{C} \times \mathbb{C}$ into $\mathbb{C}$ taking (z, w) into $z + w$ and taking (z, w) into zw . Show that they are continuous. Show that the map of $\mathbb{C} - \{0\}$ into itself taking z into z^{-1} is continuous. Conclude the standard continuity and limit theorems for sums, products, and quotients of complex-valued functions and sequences [cf. §6, this chapter, for the remarks concerning sequences].

Completeness of a metric space. We need hardly note that the term “complete” has been employed before. It is not our object to overturn established usage. The context should make the meaning clear. A metric space X is said to be *complete* provided that every Cauchy sequence possesses a limit, a sequence $(a_n): \mathbb{N} \rightarrow X$ being termed *Cauchy* provided that, as one would expect, for each positive η there exists $m \in \mathbb{N}$ such that $\rho(a_p, a_q) < \eta$ when $p, q \geq m$. Clearly, $\mathbb{R}$ and $\mathbb{C}$ are complete in the sense of their standard metric.

Sequences of maps. Let X be a given set and let (Y, ρ) be a metric space. Let $f, f_n: X \rightarrow Y, n \in \mathbb{N}$. The definitions of §3, Ch. II concerning pointwise and uniform limits and pointwise and uniform Cauchy sequences are to be paraphrased with $\rho(a, b)$ replacing $|b - a|$ (and that concerning $\varepsilon(n)$ when $X \neq \emptyset$). The notion of completeness plays the role of guarantor: when Y is a complete metric space, a pointwise (resp., uniform) Cauchy sequence possesses a pointwise (resp., uniform) limit.

EXERCISES

5.2. Prove the last assertion.

5.3. Let X be a topological space, $a \in X$. It is supposed that (f_n) tends uniformly to f on X and that each f_n is continuous at a . Show that f is continuous at a . Hint: Paraphrase the usual argument of elementary real analysis, starting with

$$\begin{aligned} \rho[f(a), f(x)] &\leq \rho[f(a), f_n(a)] + \rho[f_n(a), f_n(x)] + \rho[f_n(x), f(x)] \\ &\leq 2\varepsilon(n) + \rho[f_n(a), f_n(x)] \end{aligned}$$

for n large.

We shall have frequent occasion to consider the case where X is a Hausdorff space (often an open set in $\mathbb{C}$) and to treat sequences (f_n) satisfying

either (1) for each compact set $A \subset X$ the sequence $(f_n | A)$ possesses a uniform limit or (2) for each compact $A \subset X$ the sequence $(f_n | A)$ is a uniform Cauchy sequence. Following the terminology of Valiron we say that (f_n) possesses a uniform limit IN X provided that (1) holds and that (f_n) is a uniform Cauchy sequence IN X provided that (2) holds.

Uniform continuity. Suppose that (X, ρ) and (Y, μ) are metric spaces. Let $f: X \rightarrow Y$. The notion of uniform continuity paraphrases the classical version. We say that f is *uniformly continuous* provided that for each positive η there exists a positive δ such that

$$\mu[f(x), f(y)] < \eta$$

when $\rho(x, y) < \delta$. In the nontrivial case where $X \neq \emptyset$, there is a very elegant means of expressing the notion of uniform continuity due to de la Vallée Poussin. For $\delta \geq 0$ we consider

$$\{\mu[f(x), f(y) : \rho(x, y)] \leq \delta\}. \quad (5.1)$$

When this set is bounded, we denote its supremum by $\omega_f(\delta)$. The set of δ for which (5.1) is bounded is a connected subset of $\mathbf{R}^+$ containing 0. The function ω_f assigning to each such δ the value $\omega_f(\delta)$ is termed the *modulus of continuity* of f . Clearly ω_f is never decreasing, that is, $\omega_f(a) \leq \omega_f(b)$ for a, b in the domain of ω_f satisfying $a < b$. The relation between uniform continuity and modulus of continuity is given in Ex. 5.4.

EXERCISES

5.4. f is uniformly continuous if and only if the domain of ω_f does not reduce to 0 and ω_f is continuous at 0. It is supposed that $X \neq \emptyset$.

5.5. Determine ω_f in the following cases: $X = [0, 1], f(x) = x$; $X = \mathbf{R}^+, f(x) = \sqrt{x}$; $X = [0, 1], f(x) = x^2$; $X = \mathbf{R}^+, f(x) = x^2$.

5.6. Given $f: [0, 1] \rightarrow Y$ where Y is a metric space. If $\lim_{\delta \rightarrow 0} \delta^{-1} \omega_f(\delta) = 0$, then f is constant.

5.7. Given $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are both uniformly continuous, $X \neq \emptyset$. Show that with $h = g \circ f$ for δ small we have $\omega_h(\delta) \leq \omega_g[\omega_f(\delta)]$.

Important because of its applications is

Theorem 5.1: Let $f: X \rightarrow Y$ where X is a compact metric space, Y is a metric space, and f is continuous on X . Then f is uniformly continuous.

PROOF: We put aside the trivial case $X = \emptyset$. Let η be positive. For each $x \in X$ let $\delta(x)$ be positive and such that $\mu[f(x), f(y)] < \eta/2$ when

$\rho(x, y) < 2\delta(x)$, ρ and μ being the metrics of X and Y , respectively. The family of balls $B[x; \delta(x)]$, $x \in X$, is an open covering of X . By the compactness of X there exists a finite subfamily

$$\{B[x_1; \delta(x_1)], \dots, B[x_n; \delta(x_n)]\}$$

with union X . Given y, z such that

$$\rho(y, z) < \delta = \min\{\delta(x_1), \dots, \delta(x_n)\},$$

we see that $y \in B[x_j; \delta(x_j)]$ for some $j \in \langle 1, n \rangle$, say k . By the triangle inequality

$$\rho(x_k, z) \leq \rho(x_k, y) + \rho(y, z) < 2\delta(x_k),$$

so that $\mu[f(x_k), f(z)] < \eta/2$. Since $\mu[f(x_k), f(y)] < \eta/2$, we conclude that $\mu[f(y), f(z)] < \eta$. The uniform continuity of f is established.

EXERCISES

5.8. Let A denote a compact subset of $\mathbb{C}$, $A \neq \emptyset$, and F a closed subset of $\mathbb{C}$, $F \neq \emptyset$. Then the function f given by $f(z, w) = |z - w|$, $(z, w) \in A \times F$, has a minimum. Show that this is not true when "compact" is replaced by "closed."

5.9. A compact metric space X is sequentially compact in the sense that given a sequence $(x_n)_n^\infty$ of points of X , there exists a subsequence $(x_{n(n)})$ possessing a limit in X . Prove this assertion by establishing the existence of a point $a \in X$ such that for each positive r the set

$$\{n: x_n \in B(a; r)\}$$

is unbounded.

Prove Theorem 5.1 with the aid of the sequential compactness property.

5.10. Let (X, ρ) and (Y, σ) be metric spaces. Show that

$$\tau[(x_1, y_1), (x_2, y_2)] = \rho(x_1, x_2) + \sigma(y_1, y_2)$$

defines a metric on $X \times Y$ which induces the product topology on $X \times Y$.

Show that if X and Y are compact and f is a continuous map of $X \times Y$ into a metric space Z , then f is uniformly continuous in the sense of the metric τ . It is to be noted that this result extends to products $X_1 \times \dots \times X_n$ of compact metric spaces $X_1, \dots, X_n$.

6. COMPACTIFICATION

Given a topological space X , a topological space Y will be termed a *compactification* of X provided that (1) Y is a compact topological space, (2) $X \subset Y$, (3) the given topology of X is its relative topology with respect to Y . It is clear that if X admits a compactification Y , then $\bar{X}$ is also a compactification, closure being taken in the sense of Y . A necessary condition

for a topological space X to admit a compactification in which it is open is that X be Hausdorff and that for each $x \in X$ there exists a compact neighborhood V (neighborhood in the sense of the topology on X so that $V \subset X$). The converse is true. We put aside the trivial case where X is compact and show: *If X is a noncompact Hausdorff space with the property that every point possesses a compact neighborhood, then there exists a compactification consisting of X and a point $\infty \notin X$.* Such a compactification is termed an *Alexandroff* (or *one-point*) compactification.

To establish the assertion, we take as a clue the observation that if Y is a compactification of X and A is a compact subset of X , then $Y - A$ is open. We propose $Y = X \cup \{\infty\}$, where ∞ is an element not in X and as a candidate topology for Y the union of the topology on X and the collection of sets $Y - A$, A compact $\subset X$. It serves.

EXERCISE

6.1. Verify the last assertion. Show also that any two Alexandroff compactifications of X are homeomorphic.

It is appropriate to ask what this means for $\mathbb{C}$ and complex function theory. To be sure, we are not dealing with the indispensable, but rather the convenient and the illuminating. As we see at once, $\mathbb{C}$ admits an Alexandroff compactification. We denote the adjoined element by ∞ (infinity) and the compactification by $\hat{\mathbb{C}}$. We term $\hat{\mathbb{C}}$ the *extended complex plane*. Limit statements for $1/z$ now have a concrete meaning. Circumlocutions go into precise statements. We define $1/0$ (also denoted 0^{-1}) to be ∞ and $1/\infty$ (also denoted ∞^{-1}) to be 0 . We now see that the map of $\hat{\mathbb{C}}$ into itself taking z into $1/z$ is a homeomorphism of $\hat{\mathbb{C}}$ onto itself. Hence, if f is a continuous map of a topological space X into $\hat{\mathbb{C}}$, then so is $g \circ f$, where $g(z) = 1/z$, $z \in \hat{\mathbb{C}}$. We shall see later that it is very convenient to allow functions taking the value ∞ , especially in the study of meromorphic functions. In general, if f is a map with image in $\hat{\mathbb{C}}$, we denote $g \circ f$ by $1/f$.

EXERCISE

6.2. Relate for $(a_n): \mathbb{N} \rightarrow \mathbb{C}$ the statement " $\lim a_n = \infty$ " of §3, Ch. II and " $\lim a_n = \infty$ " in the sense of the topology on $\hat{\mathbb{C}}$.

We introduce by agreement various operations involving ∞ . Thus when $a \neq 0$, we define $a\infty$ and $a/0$ to be ∞ ; when $a \in \mathbb{C}$, we define a/∞ to be 0

and $\infty + a$, $a + \infty$, $\infty - a$, and $a - \infty$ to be ∞ . Thereby f given by az (resp., a/z) is continuous on $\hat{\mathbb{C}}$ when $a \in \mathbb{C} - \{0\}$, and f given by $a + z$ is continuous on $\hat{\mathbb{C}}$ when $a \in \mathbb{C}$.

The extended real line. For the purposes of real analysis we wish to distinguish the “large positive” and the “large negative.” To that end we introduce two distinct elements, neither in $\mathbb{R}$, designated $-\infty$ and $+\infty$, respectively. We define $\hat{\mathbb{R}}$ as $\mathbb{R} \cup \{-\infty, +\infty\}$ and proceed by first defining an order $<$ on $\hat{\mathbb{R}}$ which agrees with that on $\mathbb{R}$ for elements in $\mathbb{R}$ and thereupon introducing a topology on $\hat{\mathbb{R}}$ rendering $\hat{\mathbb{R}}$ a compactification of $\mathbb{R}$. Given $a, b \in \hat{\mathbb{R}}$, we say that $a < b$ if and only if one of the following holds: (1) $a = -\infty$, $b \neq -\infty$, (2) $a \in \mathbb{R}$ and either $b = +\infty$ or $b \in \mathbb{R}$ and $b - a$ is positive. We propose as a topology T for $\hat{\mathbb{R}}$ the collection of unions of sets of the following kinds: $]a, b[$, $\{-\infty \leq x < a\}$, $\{a < x \leq +\infty\}$, where $a, b \in \mathbb{R}$, $a < b$. T serves.

EXERCISE

6.3. Verify the last assertion.

We term $\hat{\mathbb{R}}$ the *extended real line*. We introduce “ $]$ ” and “[$]$ ” as for $\mathbb{R}$ but now with reference to $\hat{\mathbb{R}}$. Every subset of $\hat{\mathbb{R}}$ has a supremum and an infimum defined in terms of the order on $\hat{\mathbb{R}}$. It will be convenient to write $\sup_A f$ for the supremum of the image of $f|_A$ where $f: X \rightarrow \hat{\mathbb{R}}$ and $A \subset X$, similar agreements holding for $\inf_A f$ and, when meaningful, for $\max_A f$ and $\min_A f$.

We define $(+\infty) + a$ and $a + (+\infty)$ to be $+\infty$ when $a > -\infty$; $(-\infty) + a$ and $a + (-\infty)$ to be $-\infty$ when $a < +\infty$; $(+\infty)a$ and $a(+\infty)$ to be $+\infty$ when $a > 0$ and $-\infty$ when $a < 0$; $(-\infty)a$ and $a(-\infty)$ to be $-\infty$ when $a > 0$ and $+\infty$ when $a < 0$. We define $|\infty|$, $|+\infty|$, and $|\infty|$ to be $+\infty$. Finally, we extend the definition of a^+ as $\max\{a, 0\}$ and a^- as $\min\{a, 0\}$ to $a \in \hat{\mathbb{R}}$.

Given $f, g: X \rightarrow \hat{\mathbb{R}}$, by “ $f < g$ ” is meant “ $f(x) < g(x)$, $x \in X$ ”; $\leq$, $>$, and $\geq$ are similarly defined.

EXERCISES

6.4. Show that if $A \subset \mathbb{R}$ and A is nonempty and bounded above, then $\sup A$ in the sense of $\mathbb{R}$ is the same as $\sup A$ in the sense of $\hat{\mathbb{R}}$, and that a similar observation holds relative to the notion of infimum.

6.5. Subsume the definition of the limit of a sequence under that of a function using the relative topology of $\mathbb{N}$ considered as a subset of $\hat{\mathbb{R}}$.

6.6. Paraphrase the developments leading to the notions of limit superior (inferior) for $(a_n): \mathbf{N} \rightarrow \hat{\mathbf{R}}$, noting that here “sup” and “inf” exist. Compare for $(a_n): \mathbf{N} \rightarrow \mathbf{R}$ the $\mathbf{R}$ -theory and the $\hat{\mathbf{R}}$ -theory.

6.7. Return to the other material of Chs. I and II in which the symbols $+\infty$ and $-\infty$ were used and reinterpret the developments in question in terms of the language of $\hat{\mathbf{R}}$ (in particular, the Cauchy–Hadamard theorem, framing the appropriate conventions for the reciprocal on $[0, +\infty]$).

6.8. Let $a, b \in \mathbf{C}$. Let $0 \leq r_1 < r_2 \leq +\infty$ and let $0 \leq s_1 < s_2 \leq +\infty$. Show that if

$$\{z: r_1 < |z - a| < r_2\} = \{z: s_1 < |z - b| < s_2\},$$

then $(a, r_1, r_2) = (b, s_1, s_2)$.

6.9. Given $(a_n): \mathbf{N} \rightarrow \mathbf{C}$, let $b_n = |a_{n+1}|^{-1/(n+1)}$ if $a_{n+1} \neq 0$, and let $b_n = +\infty$ if $a_{n+1} = 0$, $n \in \mathbf{N}$. Develop the Cauchy–Hadamard theorem in the following form: the radius of convergence of $\sum_0^\infty a_k z^k$ is $\liminf_{n \rightarrow \infty} b_n$, giving a proof starting with the sequence (b_n) .

This is a convenient moment to furnish a gloss for modes of expression such as (1) “ $P(x)$ when x is (sufficiently) *near* a ,” (2) “ $P(x)$ when x is *small*,” (3) “ $P(x)$ when x is *large*.” We shall understand for (1) “there exists a neighborhood U of a such that $P(x)$ holds when $x \in U$.” For “ x is small” we construe “ x is near 0” and for “ x is large” we construe “ $1/x$ is small,” reference being made to x in a subset of $\hat{\mathbf{C}}$. For “ $P(x)$ when x is *near but different from* a ” we shall understand the gloss for (1) with “ $x \in U$ ” replaced by “ $x \in U$, $x \neq a$.” When we say “ $f(x)$ is (arbitrarily) near b when x is (sufficiently) near a ,” we mean that $f \cup \{(a, b)\}$ is a function continuous at a . When we say “ $f(x)$ is near b when x is near but different from a ,” we mean that the assertion glossed in the preceding sentence holds with f replaced by the restriction of f to its domain less $\{a\}$. “Near a ” and “near b ” in these assertions will be replaced on occasion by “large,” “small,” the first meaning “near ∞ ,” the second “near 0.”